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ORIGINAL WORK

Dysautonomia in Guillain–Barré Syndrome: Prevalence, Clinical Spectrum, and Outcomes

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Abstract

Background: Guillain–Barré syndrome (GBS), when severe, involves the autonomic nervous system; our objective was to assess the spectrum and predictors of dysautonomia, and how it may impact functional outcomes.

Methods: A retrospective review of patients admitted to the Mayo Clinic in Rochester, MN between January 1, 2000, and December 31, 2017, with GBS and dysautonomia was performed. Demographics, comorbidities, parameters of dysautonomia, clinical course, GBS disability score, and Erasmus GBS Outcome Score (EGOS) at discharge were recorded.

Results: One hundred eighty seven patients were included with 71 (38%) noted to have at least one manifestation of dysautonomia. There are 72% of patients with a demyelinating form of GBS and 36% of patients with demyelination had dysautonomia. Ileus (42%), hypertension (39%), hypotension (37%), fever (29%), tachycardia or bradycardia (27%), and urinary retention (24%) were the most common features. Quadriparesis, bulbar and neck flexor weakness, and mechanical ventilation were associated with autonomic dysfunction. Patients with dysautonomia more commonly had cardiogenic complications, syndrome of inappropriate antidiuretic hormone, posterior reversible encephalopathy syndrome, and higher GBS disability score and EGOS. Mortality was 6% in patients with dysautonomia versus 2% in the entire cohort ($P = 0.02$).

Conclusions: Dysautonomia in GBS is a manifestation of more severe involvement of the peripheral nervous system. Accordingly, mortality and functional outcomes are worse. There is a need to investigate if more aggressive treatment is warranted in this category of GBS.

Keywords: Dysautonomia, Guillain–Barré syndrome, Acute inflammatory demyelinating polyradiculoneuropathy, Autonomic dysfunction

Introduction

The most critical features of Guillain–Barré syndrome (GBS) are involvement of the respiratory and oropharyngeal musculature, and autonomic nervous system. Signs of dysautonomia include extreme blood pressure fluctuations, cardiac arrhythmias, ileus, and urinary retention and usually resolve in the plateau phase of GBS preceding motor improvement [1]. The mechanism is thought to be an imbalance in sympatho-vagal transmission shifted toward sympathetic predominance [2] mediated by afferent fibers from arterial baroreceptors, efferent parasympathetic fibers that innervate the heart, and in preganglionic sympathetic fibers that control sudomotor and vasomotor functions [3, 4]. It is also hypothesized that afferent conduction block may induce a catecholamine surge that can lead to inappropriate secretion of antidiuretic hormone (SIADH) and renin, as well as posterior reversible encephalopathy syndrome (PRES) [5–7].

One of the earlier observations by Lichtenfeld in 1971 attributed fatality in GBS to dysautonomia; four of 28 patients died "during or immediately after

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episodes of severe autonomic dysfunction" as a result of "cardiac arrest following several hours of rapidly fluctuating autonomic status" or were "found dead after extremely high blood pressure recordings although paralysis was not severe." Dysautonomia in GBS was subsequently described in up to two-thirds of patients in older series [8–11], but it is unclear what its frequency is in current practice. The objective of this study was to assess the contemporary prevalence of dysautonomia in patients with GBS, and the spectrum and relative frequency of different manifestations of autonomic dysfunction. We also investigated what GBS factors are associated with dysautonomia and how this impacts functional outcomes.

Methods

Patient Population and Study Design

The electronic medical record was searched for patients admitted to the Mayo Clinic in Rochester, MN between January 1, 2000, and December 31, 2017, with a billable diagnosis of GBS. Demographics, comorbidities, features of GBS including nerve conduction studies, clinical course, hospital complications, and functional outcomes were recorded. Pertinent comorbidities assessed in the medical history were those that can cause neuropathic symptoms including diabetes mellitus, chronic neuropathy, alcoholism, and "other," encompassing a history of malignancy, chemotherapy or immunotherapy, multiple sclerosis, or metabolic disorder other than diabetes such as hyperaldosteronism.

Patients with subsequent chronic inflammatory polyradiculoneuropathy (CIDP) were included in the study. Specific information regarding antibody status (i.e., anti-GQ1b or anti-nicotinic antibodies) was not readily available mainly due to lack of consistent testing and/or reporting of results in paper charts versus the electronic medical record. Patients were excluded if they did not have a definite diagnosis of GBS, did not have adequate data with respect to parameters of dysautonomia such as hemodynamics, or had an incomplete neurologic examination. Records were additionally assessed for the presence or absence of autonomic dysfunction. The diagnosis of dysautonomia was made either based off of billing diagnoses as documented, or by individual chart assessment of vitals and refractory severe hypertension, wide fluctuations in blood pressure, ileus, urinary retention, sudomotor or pupillary dysfunction, or fever or hypothermia not attributed to other causes such as infection [1, 2]. Patients with one or more of these signs were deemed as having dysautonomia.

Nerve Conduction Studies

GBS type was differentiated as axonal, demyelinating, both axonal and demyelinating, or "other" such as Miller–Fisher syndrome based on nerve conduction studies as interpreted by a neurophysiologist. Nerve conduction findings were further assessed by recording summated distal compound muscle action potential (CMAP) amplitude in millivolts (mV), and whether this amplitude was less than 20% of the lower limit of normal amplitudes based on the motor study performed. The number of inexcitable nerves in the study was also recorded.

Neurological Course

The presence of neuropathic pain based on billable diagnoses as well as individual chart review of patient symptoms in the hospital admission was recorded. Motor weakness on physical examination performed by a neurologist was classified as mild, paraparetic, quadriparetic, bulbar, or involving neck flexion. Sensory loss was categorized as mild (focal) or complete (diffuse and extensive). Intubation and mechanical ventilation were recorded. As a measure of GBS severity, we calculated the Erasmus GBS Outcome Score (EGOS) [11].

Features of Dysautonomia

Parameters of dysautonomia recorded were tachycardia > 120 beats per minute or tachycardia > 100 beats per minute with beta-blocker use lasting over 2 h, bradycardia < 50 beats per minute for over 2 h, highest and lowest heart rate, hypertension with systolic blood pressure > 180 or > 160 mmHg on antihypertensive medications lasting over 2 h, maximum systolic blood pressure, minimum diastolic blood pressure, hypotension with systolic blood pressure < 90 mmHg, and longest durations of these abnormalities. Whether or not patients were treated for the particular parameter with antihypertensives, vasopressors, or rate control agents was recorded. Other features of dysautonomia noted were ileus or gastric dysmotility (either unequivocally diagnosed by radiology or requiring pharmacological treatment or mechanical decompression), urinary retention not caused by obstruction and requiring catheterization, sudomotor dysfunction (hyperhidrosis), fever > 38.3 °C not attributable to superimposed infection [12], and hypothermia < 36 °C. Treatments such as bowel decompression were documented.

Treatment and Complications

GBS treatment including the use of intravenous immunoglobulin (IVIG), plasmapheresis (PLEX), or both was noted. Additionally, hospital complications recorded were sepsis, cardiogenic consequences such as arrhythmias whether or not related to dysautonomia, pulmonary embolism or deep venous thrombus, substance withdrawal, and “other” including pneumonia, urinary tract infection, malignancy or paraneoplastic disorder, or disseminated intravascular coagulation. The presence of SIADH was noted based on serum sodium and urine studies. PRES was defined by clinical and radiological data [13] not attributable to medications. Whether PRES preceded GBS and dysautonomia, was in the context of dysautonomia, or occurred after IVIG was also assessed.

Endpoints

Study outcomes were length of intensive care unit (ICU) and hospital stays, whether patients were ambulatory upon discharge, the GBS disability score [14] at discharge, as well as mortality and disposition (home or skilled nursing facility) at discharge.

Statistical Analysis

Qualitative data were described as nominal or ordinal variables, and as mean and standard deviation or median and interquartile range (IQR) for continuous variables. Data were analyzed using JMP 10.0.0 software (SAS Institute, Cary, North Carolina). We used a t test or Chi-square test (or Fisher’s exact test for $n<5$) to assess associations amongst variables. Odds ratios (OR) and 95% confidence intervals (CI) were computed for nominal data. Statistical testing was done at the two-tailed α level of .05.

Results

One hundred eighty seven patients were included in the study, of which 71 (38%) were noted to have signs of dysautonomia. Demographics, clinical course, nerve conduction studies, and cerebrospinal fluid results are shown in Table 1. Clinical features of patients with dysautonomia are shown in Table 2. Treatments, hospital complications, outcomes, and corresponding statistical analyses are shown in Table 3. Men were more prevalent in the entire cohort (59%) and in the cohort with dysautonomia (43/71, 61%). Sex and age were not associated with dysautonomia. Diarrhea preceded GBS in 35/187 (19%) patients. Recurrence of GBS evolving into CIDP was seen in 9%, and recurrence was not a predictor of autonomic instability.

One sign of autonomic instability was present in 47/71 (66%) patients. Sustained tachycardia was more common than bradycardia (26% versus 1%), while hypertension (39%) and hypotension (37%) were similarly prevalent. Fevers (29%) were more common than hypothermia (6%). Ileus was the most common manifestation of dysautonomia (present in 42%). Urinary retention was present in 24%. 10% patients had sudomotor dysfunction, and 14% had pupillary dysfunction. Four (6%) patients had orthostatic hypotension.

Most patients had a demyelinating form of GBS (72%), including 48/134 (36%) who had dysautonomia. 11/22 (50%) patients who had an axonal form had dysautonomia. GBS type (demyelinating versus axonal) was not predictive of autonomic dysfunction. Patients with dysautonomia had lower distal CMAP amplitudes, and a higher likelihood of summated distal CMAP amplitudes < 20% of the lower limit of normal. Neurologic examination findings predictive of dysautonomia were quadriparesis, bulbar weakness, and neck flexor weakness. Patients with mild weakness were less likely to have autonomic instability. Mechanical ventilation was also associated with dysautonomia, whereas neuropathic pain was not. Patients in the dysautonomia group had a higher rate of treatment with PLEX or both IVIG and PLEX. In these 21 patients, GBS score was median 4, IQR 3–4 and EGOS was median 5, IQR 4–5. CSF inflammation was not predictive of dysautonomia.

Overall, occurrence of one or more major hospital complications was associated with dysautonomia. Patients with dysautonomia more often had cardiogenic complications that were not necessarily due to autonomic instability, as well as SIADH. Four patients had minor arrhythmias; three had non-sustained ventricular tachycardia, and one had an episode of atrial fibrillation with rapid ventricular rate. Two patients had cardiac arrest after having respiratory arrest; in one case, the patient had an acute blood clot in their tracheostomy tube, causing respiratory arrest then pulseless electrical activity (PEA) arrest. In the other case, the patient had PEA arrest after respiratory arrest that occurred prior to neurologic manifestations of GBS developed. No patients required a pacemaker associated with complications due to GBS. PRES was more prevalent in patients with dysautonomia; six patients had PRES, five of them had dysautonomia, and four of them had hypertension. Three of the six patients had PRES as an initial presentation prior to developing neurologic signs of GBS (Fig. 1), and two patients had PRES after GBS were diagnosed (Fig. 2). The patient who did not have dysautonomia had PRES after IVIG treatment and was on tacrolimus following a previous cardiac transplant. Three of the patients with PRES had encephalopathy, and two had seizures including one with status epilepticus. No patients with PRES had cortical blindness.

The length of hospital stay was longer in patients with dysautonomia, and fewer of these patients were ambulatory at discharge. The GBS disability score and EGOS were both significantly higher in those with autonomic instability. Patients with autonomic instability were

Table 1 Patient demographics and clinical features

Characteristic	All $N=187$	Dysautonomia $N=71$	No dysautonomia $N=116$	P value	OR (CI)
Female	76 (40.6)	28 (39.4)	48 (41.4)	0.79	0.92 (0.50–1.69)
Age (y)	55 ± 16	55 ± 16	54 ± 16	0.82	–
Comorbidities					
Diabetes mellitus II	22 (11.8)	7 (9.9)	15 (12.9)	0.53	0.74 (0.28–1.90)
Neuropathy	11 (5.9)	4 (5.6)	7 (6.0)	1.00	0.93 (0.26–3.30)
Alcoholism	2 (1.1)	2 (2.8)	0 (0.0)	0.07	–
Other[a]	41 (21.9)	15 (21.1)	26 (22.4)	0.84	0.93 (0.45–1.90)
CIDP	17 (8.9)	9 (12.7)	8 (6.9)	0.18	1.96 (0.72–5.33)
GBS type					
Axonal	22 (11.8)	11 (15.5)	11 (9.5)	0.22	1.75 (0.72–4.28)
Demyelinating	134 (71.7)	48 (67.6)	86 (74.1)	0.34	0.73 (0.38–1.39)
Both	21 (11.2)	8 (11.3)	13 (11.2)	0.99	1.01 (0.40–2.56)
Other (MF)	10 (5.3)	4 (5.6)	6 (5.2)	1.00	1.09 (0.30–4.02)
NCS[b]					
Summated distal CMAP amplitudes (mV)	10.4 ± 8.6	8.1 ± 7.9	11.8 ± 8.8	0.004	–
Summated distal CMAP amplitudes < 20% LLN	45 (23.4)	27/68 (39.7)	18/111 (16.2)	< 0.001	3.40 (1.69–6.86)
# Inexcitable nerves	0, IQR 0–0	0, IQR 0–0	0, IQR 0–0	0.10	–
Motor weakness	171 (91.4)	67 (94.3)	104 (89.7)	0.30	1.93 (0.60–6.24)
Mild	86 (46.0)	23 (32.4)	63 (54.3)	0.004	0.40 (0.22–0.75)
Paraparetic	28 (15.0)	12 (16.9)	16 (13.8)	0.56	1.27 (0.56–2.87)
Quadriparetic	57 (30.5)	32 (45.1)	25 (21.6)	< 0.001	2.99 (1.57–5.69)
Bulbar	69 (36.9)	37 (52.1)	32 (27.6)	< 0.001	2.86 (1.54–5.30)
Neck flexor	48 (25.7)	31 (43.7)	17 (14.7)	< 0.001	4.51 (2.25–9.05)
Sensory loss	122 (65.2)	47 (66.2)	75 (64.7)	0.83	1.07 (0.57–1.99)
Mild	115 (6.1)	42 (59.2)	73 (62.9)	0.60	0.85 (0.47–1.56)
Complete	7 (3.7)	5 (7.0)	2 (1.7)	0.12	4.32 (0.81–22.88)
Neuropathic pain	94 (50.3)	39 (54.9)	55 (47.4)	0.32	1.35 (0.74–2.45)
Mechanical ventilation	51 (27.2)	36 (50.7)	15 (12.9)	< 0.001	6.93 (3.39–14.15)
CSF protein	83, IQR 54–135	78, IQR 54–119	90, IQR 52–142	0.46	–
CSF leukocytes	2, IQR 1–4	1, IQR 1–6	2, IQR 1–4	0.40	–

Data presented as *N* (%), mean ± standard deviation, or median, IQR

CI confidence interval, *CIDP* chronic inflammatory demyelinating polyneuropathy, *CMAP* compound muscle action potential, *CSF* cerebrospinal fluid, *d* days, *GBS* Guillain–Barré syndrome, *IQR* interquartile range, *LLN* lower limit of normal, *MF* Miller–Fisher, *mV* millivolts, *N* sample size, *NCS* nerve conduction studies, *OR* odds ratio, *y* years

[a] Malignancy, chemotherapy, multiple sclerosis, immunotherapy, or metabolic disorder other than diabetes such as hyperaldosteronism

[b] 179 patients had nerve conduction studies

less likely to be discharged home and more likely to go to a skilled nursing facility. Overall mortality at hospital discharge was 2% and higher in those with autonomic dysfunction (6%). All four patients who died had dysautonomia at some point in their hospital course, but none died as a direct consequence of the autonomic instability. Two patients died with lymphoma with subsequent withdrawal of life-sustaining interventions, and another patient also had life-sustaining interventions withdrawn after developing multiorgan failure and coma of unknown etiology, and the fourth patient died of brain herniation after developing an acute on chronic subdural hematoma. Two patients with lymphoma had CSF cytology studies, and one had a sural nerve biopsy that did not reveal lymphoma; both had EMGs with features suggestive of demyelination.

Discussion

Dysautonomia and organ dysfunction play no small part in the syndrome of severely affected patients with GBS and can be immediately life-threatening. The rate of autonomic dysfunction in our study was high at 38% with a mortality rate of 6%. Patients with dysautonomia also had a longer length of hospital stay, were less prone to be

Table 2 Features of dysautonomia

Clinical feature	N	Median value of extremes[a]	IQR[a]	Longest duration (h)	Treatment[a]
Tachycardia > 2 h	18 (26.1)	123 (rate)	110–140	3.0 ± 9.2	11/18 (61.1)
Bradycardia > 2 h	1 (1.4)	59 (rate)	49–66	0.2 ± 1.0	0
Hypertension > 2 h	27 (38.6)	190 (SBP)	170–215	3.9 ± 12.3	3/27 (11.1)
Hypotension > 2 h	26 (37.1)	49 (DBP)	37–57	1.1 ± 5.8	12/26 (46.2)
Ileus	30 (42.3)	–	–	–	11/30 (36.7)
Urinary retention	17 (23.9)	–	–	–	–
Sudomotor dysfunction	7 (9.9)	–	–	–	–
Pupillary dysfunction	10 (14.1)	–	–	–	–
Fever > 38.3	20 (28.6)	–	–	–	–
Hypothermia < 36	4 (5.7)	–	–	–	–

Data presented as N (%), median, or IQR

DBP diastolic blood pressure, *GBS* Guillain–Barré syndrome, *h* hours, *hrs* hours, IQR interquartile range, *N*, sample size, *SBP* systolic blood pressure

[a] For any hemodynamic instability (not just sustained)

Table 3 Treatments, complications, and outcomes

Characteristic	All *N* = 187	Dysautonomia *N* = 71	No dysautonomia *N* = 116	*P* value	OR (CI)
GBS treatment	179 (95.7)	69 (97.2)	110 (94.8)	0.71	1.88 (0.37–9.59)
IVIG	146 (78.1)	48 (67.6)	98 (84.5)	0.007	0.38 (0.19–0.77)
PLEX	17 (9.1)	10 (14.1)	7 (6.0)	0.06	2.55 (0.92–7.05)
Both	16 (8.6)	11 (15.5)	5 (4.3)	0.01	4.07 (1.35–12.26)
Hospital complications	32 (17.1)	22 (31.0)	10 (8.6)	< 0.001	4.76 (2.09–10.81)
Sepsis	6 (3.2)	3 (4.2)	3 (2.6)	0.68	1.66 (0.33–8.47)
Cardiogenic	7 (3.7)	6 (8.5)	1 (0.9)	0.01	10.62 (1.25–90.11)
PE/DVT	7 (3.7)	4 (5.6)	3 (2.6)	0.43	2.25 (0.49–10.36)
Substance withdrawal	3 (1.6)	2 (2.8)	1 (0.9)	0.56	3.33 (0.30–37.45)
SIADH	27 (14.4)	17 (23.9)	10 (8.6)	0.004	3.34 (1.43–7.78)
PRES	6 (3.2)	5 (7.0)	1 (0.9)	0.03	8.71 (0.99–76.17)
Other[a]	18 (9.6)	14 (19.7)	4 (3.5)	0.001	6.88 (2.16–21.85)
Outcomes					
ICU admission	83 (44.4)	54 (76.1)	29 (25.0)	< 0.001	9.53 (4.79–18.97)
ICU stay (d)	13 ± 13	14 ± 14	9 ± 10	0.07	–
Hospital stay (d)	20 ± 56	33 ± 88	12 ± 14	< 0.001	–
Ambulatory	108 (57.8)	30 (42.3)	78 (67.2)	< 0.001	OR 0.36 (0.19–0.66)
GBS disability score	3, IQR 2–4	4, IQR 3–4	3, IQR 2–4	< 0.001	–
	3.0 ± 1.2	3.6 ± 1.2	2.7 ± 1.1		
EGOS	4, IQR 3–5	5, IQR 4–5	4, IQR 3–5	<0.001	–
	3.8 ± 1.3	4.3 ± 1.3	3.5 ± 1.2		
Home	74 (39.6)	16 (22.5)	58 (50.0)	< 0.001	OR 0.29 (0.15–0.57)
SNF	109 (58.3)	51 (71.8)	58 (50.0)	0.003	OR 2.55 (1.35–4.80)
Mortality	4 (2.1)	4 (5.6)	0 (0.0)	0.02	–

Data presented as *N* (%), mean ± standard deviation, or median, IQR

CI confidence interval, *d* days, *EGOS*, Erasmus GBS Outcome Score, *GBS* Guillain–Barré syndrome, *ICU* intensive care unit, *IQR* interquartile range, *IVIG* intravenous immunoglobulin, *N* sample size, *OR* odds ratio, *PE/DVT* pulmonary embolism or deep vein thrombus, *PLEX* plasmapheresis, *PRES* posterior reversible encephalopathy syndrome, *SIADH* syndrome of inappropriate antidiuretic hormone, *SNF* skilled nursing facility, *y* years

[a] Pneumonia, urinary tract infection, malignancy/paraneoplastic, disseminated intravascular coagulation

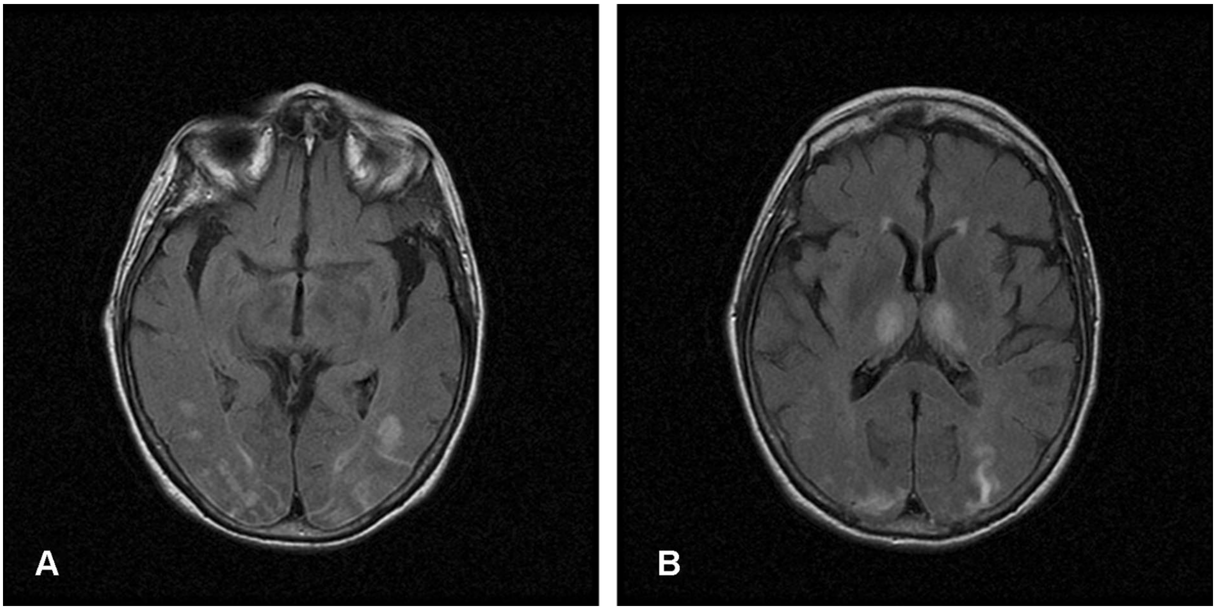

Fig. 1 Patient with PRES Prior to GBS Diagnosis. 82-year-old woman who presented with encephalopathy and status epilepticus then progressive ascending weakness in the extremities eventually diagnosed as GBS. MRI head T2 FLAIR sequences show fairly symmetric hyperintensities in the subcortical more than cortical regions of **a** bilateral parieto-occipital lobes and **b** bilateral thalami

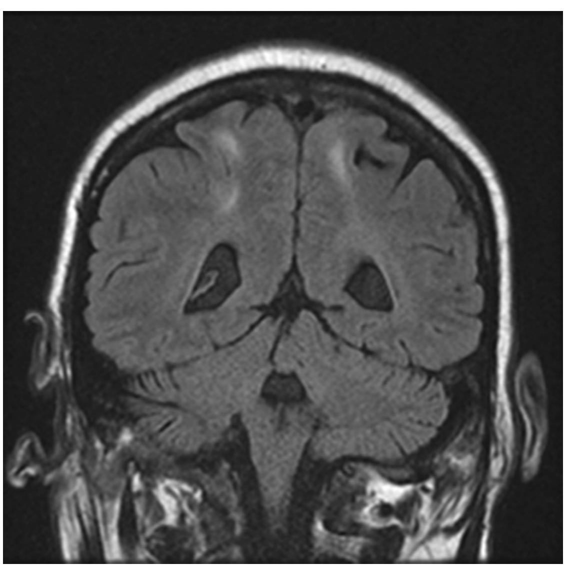

Fig. 2 Patient with PRES Following Dysautonomia with Blood Pressure Fluctuations. 51-year-old man who had GBS in the setting of diffuse large B-cell lymphoma who then had a generalized tonic–clonic seizure, then MRI with T2 FLAIR sequences showing symmetric subcortical white matter hyperintensities in bilateral parietal lobes. Preceding the seizure, he had prolonged systolic blood pressures in the 250's mm Hg alternating with systolic blood pressures in the 90's mm Hg

ambulatory at discharge, had poorer GBS disability scores and EGOS score and were more likely to be discharged to a skilled nursing facility. Moreover, mortality in patients with dysautonomia was 6% versus 0% in those without dysautonomia. Previously reported rates of dysautonomia in GBS patients have ranged from 10 to 67% [8–10, 15, 16]. Mortality rate in seminal series ranged from 6 to 13% [17–19]. The lower prevalence of dysautonomia and lower mortality in our more contemporary series likely reflects earlier recognition and treatment of GBS in current practice. GBS disability score at discharge was significantly higher in patients with autonomic instability. However, patients with dysautonomia also had more severe weakness, and we could not establish with our data whether the association of dysautonomia with worse functional status at discharge was independent of the severity of muscle weakness at nadir.

The spectrum of autonomic instability in our cohort comprised ileus in 42%, followed by hypertension in 39%, hypotension in 37%, fevers in 29%, tachycardia or bradycardia in 27%, urinary retention in 24%, pupillary dysfunction in 14%, sudomotor dysfunction in 10%, and hypothermia in 6%. Other studies have shown varying rates of different features of autonomic instability; tachy or bradyarrhythmias in up to 81%, tachycardia in up to 59%, bradycardia in 21%, arrhythmias in up to 50%, hypertension in 61%, hypotension in up to 43%, ileus in 15%, sphincter dysfunction in up to 30%, and sudomotor dysfunction in 13% [8–10, 20–24]. Differences in rates may be attributable to varying study sample sizes, as well as different cutoff values and durations for classification of cardiac abnormalities. Additionally, there may be different diagnostic criteria for ileus (for example, severe constipation due to causes other than autonomic impairment versus due to autonomic involvement) complicated by the possibility of immobility, mechanical ventilation, and narcotic medications [25], as well as inclusion of ICU versus non-ICU study populations.

Serious cardiac arrhythmias have been noted to occur in up to 50% of patients with dysautonomia [21]. No patients had serious cardiac arrhythmias directly related to GBS in our cohort, though it is difficult to say with certainty whether or not the patient who had PEA arrest after respiratory arrest prior to the development of GBS arrested because of early dysautonomia. SIADH has been noted in up to 5% patients with GBS and dysautonomia [26], whereas 24% patients in our cohort with dysautonomia had SIADH. This difference may be explained by greater recognition of this possible complication. Additionally, PRES has been previously reported as an initial presentation of GBS [27, 28]. Our series confirms that PRES can be an early manifestation of GBS with dysautonomia.

Clinical predictors of dysautonomia were quadriparesis, neck flexor or bulbar weakness, and mechanical ventilation. While it was initially thought that the severity of autonomic involvement may not be related to the degree of motor or sensory disturbance [3, 29], more recent studies have shown that the rate of dysautonomia may rise to 75% in quadriparetic patients [5]. Approximately half of the patients in our series with dysautonomia were mechanically ventilated, highlighting the need for close monitoring for signs of respiratory failure in these patients. Yet, it is pertinent to note that in our practice, we have a lower threshold to intubate patients with signs of dysautonomia as emergency intubation can be particularly dangerous in these cases. A previous study using sensitive tests of autonomic function (rather than clinical manifestations) for the definition of acute dysautonomia reported higher rates of cardio-sympathetic hyperactivity and sudomotor dysfunction in patients with demyelinating as compared to axonal acute polyradiculoneuropathy [30]. In our series, 36% patients who had a demyelinating form had dysautonomia versus 50% who had an axonal form. Though this difference was not statistically significant, the higher proportion of patients with an axonal form may be as a result of more patients with severe disease.

Early signs of dysautonomia in GBS include refractory hypertension, wide fluctuations in blood pressure and/or heart rate, hyper- or hypothermia, sudomotor or pupillary dysfunction, as well as ileus and urinary retention. More severe weakness, bulbar involvement, and respiratory failure requiring mechanical ventilation are associated with autonomic instability. Severe and rapidly progressive weakness and manifestation of dysautonomia should prompt ICU admission with continuous cardiac monitoring and blood pressure monitoring through an arterial line to prevent cerebral complications such as PRES and serious and potentially life-threatening cardiovascular complications.

The optimal treatment of dysautonomia in GBS has not been well studied. Caution when using any medications that can affect blood pressure and heart rate is advisable. Gastric and intestinal decompression and bladder catheterization are often necessary, but pharmacological interventions for treatment of gastrointestinal paralysis or urinary retention should be used conservatively because they can have adverse hemodynamic side effects. It is best to use opiates as sparingly as possible because these agents can exacerbate adynamic ileus. It is not known whether IVIG or PLEX prevents or reduces the severity of dysautonomia in GBS. Further research in this regard would be useful.

Our study has several limitations, mostly inherent to its retrospective design. Manifestations of dysautonomia were noted based on detailed review of the electronic medical and nursing records. We were unable to say with absolute certainty that specific parameters of dysautonomia were solely associated with GBS versus preexisting disease such as hypertension, hypovolemia, alcohol/opiate withdrawal, or other comorbidities. Prevalence of ileus and urinary retention may have been underestimated because milder forms might have gone untreated, bladder catheterization may have occurred before urinary retention became evident, and medications may have contributed to their occurrence. We did not have enough information to collect the MRC sum score upon admission or at nadir; consequently, we were unable to control the association of dysautonomia with disability at discharge for the severity of weakness. We could therefore not determine whether dysautonomia influences the pace or degree of functional recovery following the acute hospitalization. We were also unable to collect follow-up data on patients after hospital discharge, thus limiting a more longitudinal trajectory of patient survival and functional status.

Conclusion

Our series emphasizes that dysautonomia is seen in over one-third of patients admitted for GBS and it is more common in patients with more severe weakness. The most common signs of dysautonomia were ileus, blood pressure fluctuations, tachy or bradyarrhythmias, and urinary retention. In-hospital mortality was greater, and functional outcomes at hospital discharge were poorer in patients with autonomic instability. Dysautonomia can lead to serious secondary complications such as PRES. There is a need to investigate if more aggressive treatment is warranted in this category of GBS.

Abbreviations
GBS: Guillain–Barré syndrome; AIDP: Acute demyelinating polyradiculoneuropathy; ICU: Intensive care unit; PRES: Posterior reversible encephalopathy syndrome; SIADH: Syndrome of inappropriate antidiuretic hormone; EGOS: Erasmus GBS Outcome Score; CIDP: Chronic inflammatory polyradiculoneuropathy; CMAP: Compound muscle action potential; mV: Millivolts; CSF: Cerebrospinal fluid; IVIG: Intravenous immunoglobulin; PLEX: Plasmapheresis; PE: Pulmonary embolism; DVT: Deep vein thrombus; IQR: Interquartile range; OR: Odds ratio; CI: Confidence interval.

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Author Contributions
This study required no sponsorship or funding. Tia Chakraborty MD authored, conceptualized, acquired data, performed all statistical analyses, drafted, and revised the manuscript. She had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. Christopher Kramer MD analyzed and acquired data, conceptualized, drafted, and revised the manuscript. Eelco Wijdicks MD PhD authored, conceptualized, drafted, and revised the manuscript. Alejandro A. Rabinstein MD authored, conceptualized, drafted, and revised the manuscript.

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Conflict of interest
All authors declare that they have no conflict of interest.

Ethical Approval/Informed Consent
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